AgNW reservoir computing: where we stand (13 Oct 2026)

Question. Can a random Ag nanowire network (framework geometry and Kirchhoff circuit, unchanged) work as a reservoir computer? We design junction model, electrodes, input and readout. **G1** = computes beyond its input (MC ≥ 1, degree-2 ≥ 1, NARMA-10 ≤ 0.70; input alone 0.80); **G2** = $\geq 50\%$ of an equal-size ESN. Operating point chosen on validation data; later decision rules fixed before the runs.

What we did

- **A. Framework as is** (25-28 Sep): binary junction, one electrode pair (E1-E3); noise removed (N1, N2); averaging, multi-electrode (N5, N7); Fig. 4 filament junction (N4), slow inputs (E4).
- **B. New junction** (28 Sep-1 Oct): N10 volatile junction from measured AgNW data; 32 probes (N10-1); 16 electrodes (N10-E); memory limit (N10-M); settings (N10-S).
- **C. Realistic noise** (2-6 Oct): ruptures (N10-R); rate from Milano 2025 data (N10-C); readout noise (N10-RN); remedies: backbone (N10-K), retraining (N10-OR), bias + small input (N10-B), matched voltage scale (N10-V).
- **D. Size and layout** (7-13 Oct): 1-4 mm, 10 networks per size, HPC cluster (N10-L, N10-LS); edge bands vs pads (N10-SB).

What we found

1. **Framework junctions: no reservoir.** Scores at input-only level (MC ≤ 0.26): switching noise (binary) or a frozen hold band (Fig. 4 filament). No fix helped.
2. **N10 junction: a weak reservoir.** 16 pads: MC 3.5, degree-2 6.7, NARMA-10 0.40, sine/square 1.00 $\rightarrow$ G1, G2 loose only. Memory capped at about 4 inputs. Insensitive to our chosen settings.
3. **Ruptures break it.** Tolerance about 10^{-4} /s per junction (1 mm); failures are all-or-nothing collapses (current rerouting). Readout noise harmless. No remedy restores G1; bias + small input best without noise (MC 5.8) but most fragile.
4. **Real junctions are noisier.** Milano 2025 data imply $r^* \approx 0.005$–0.03 /s per junction, 50-300 times the tolerance.
5. **Size and layout at 10^{-2} /s.** Networks passing G1: 8, 8, 9, 6 of 10 at 1, 2, 2.83, 4 mm (about 3 in 4, no trend); pooled G1 fails everywhere, on NARMA-10. About half the capacity lost at every size; size makes devices more alike, not more successful. Edge bands: 15-30 % less capacity than pads, but more robust from 2 mm (63 % vs 49 % kept).

In one sentence: with realistic junctions a nanowire network computes beyond its input, but ruptures at the measured rate halve its capacity and make about 1 device in 4 fail; neither size (to 4 mm) nor any remedy tried removes this.

What could be improved

- **Junction model** (main weakness): built from published behaviour, not fitted to single-junction data; rupture rate inferred indirectly (mostly one size, few jumps).
- **Protocol:** input windows likely below the optimum (pads better at 1.5 $\times$ onset, 3 networks).
- **Coverage:** networks to 4 mm vs about 1 cm devices; one input encoding; no application tasks; several remedies on 3 networks only. **Packaging:** no literature positioning, journal or publication figures yet; no git.

What could be done next

Model, without new measurements (all untested so far):

- **State-dependent ruptures.** Now every junction ruptures at the same constant rate. Make the rate grow with the junction's power (thermally activated, like the existing rupture above 1 mW): hot junctions on the main paths break more. Redo N10-C with it: the same measured jumps may imply a different per-junction rate and tolerance.
- **Joint fit to the Milano 2025 data we already have.** Fit voltage scale, time constants and rupture rate together to their recordings, instead of hand-setting them (N10 switches at about 10 times their voltage).
- **More heterogeneity**: spread of switching thresholds, not only of time constants.
- **Path redundancy from the network graph**: count independent current paths between electrodes; may explain why edge bands are more robust, predict tolerance vs size, guide layout design.

Readout and analysis, on existing runs (no new simulations):

- **Collapse-proof features**: a rupture shifts the baseline current; high-pass filtered or normalised currents may remove the step (periodic retraining did not).
- **Why NARMA-10 fails**: memory or number of readout channels.

New simulations: pads at 1.5 $\times$ onset with 10 networks (pre-registered); alternating-edge layouts (c, d); backbone (N10-K) with re-tuned windows; more electrodes on larger networks; 7 mm (costly).

Paper: literature positioning, journal, figures (thesis A chosen). **Experiments:** send docs/MEASUREMENT_REQUEST.md.

Details: docs/RESULTS.md (sections 1-28), docs/NEXT_STEPS.md, reports in docs/report_2026-xx-xx/.